

DOUMBIA Laye Soma

Mechatronics — Automation, Electromechanics & Industrial Robotics

Personal details

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Address: Bamako – Kalaban Coura ACI, Mali

Portfolio:

portfolio-laye-soma-doumbia.vercel.app

Soft skills

Adaptable

Organised

Thorough

Analytical thinking

Solution-oriented

Languages

French: fluent

English: intermediate

Bambara: fluent

Education

2022 – 2026 Professional Bachelor's Degree in Electrical Engineering and Industrial Computing, Mechatronics option. Institut des Sciences Appliquées (ISA), USTTB, Bamako.

August 2021 Malian Baccalaureate, Exact Sciences track — Lycée Monseigneur Jean Marie Cissé, Kalaban Coura ACI.

Interests

Sport, music

Reading, travel

Following new technologies

Profile

Graduate in Electrical Engineering and Industrial Computing (mechatronics option), focused on intelligent systems, automation and industrial robotics.

Comfortable across the whole automated system: electrical and power circuits, sensors and actuators, PLC programming, supervision and software integration. Versatile, thorough and solution-oriented.

Technical Skills

- Automation: PLC programming, Grafcet, Ladder, control of cylinders and motors
- Electrical & electromechanical: control and power circuits, wiring, single-phase, stepper and servo motors
- Industrial robotics: manipulator arm, automated sorting cell, computer vision
- Solar energy & security: photovoltaic installation (panels, inverters, batteries), CCTV
- AI & IoT: development of AI agents (LLM), artificial intelligence applied to IoT, embedded systems
- CAD: basic level from coursework — SolidWorks, AutoCAD
- Industrial computing: supervision and HMI, Kivy MD interfaces (Python), web development (Django, React, EJS)
- Office software: Word, PowerPoint, Excel, Access — advanced

Professional Experience

Sall Pro Global – Bamako | Solar energy, electricity and security

- Installation of photovoltaic solar systems and CCTV cameras: mounting, wiring, connecting inverters and batteries, configuring recorders and remote access;
- Preventive and corrective maintenance of the installations;

Generator repair workshop – Bamako | Maintenance and repair

- Diagnosis of mechanical and electrical faults, preventive and corrective maintenance, return to service;

DoniFab (DoniLab) – Bamako | Final-year internship, 3 months

- 4-axis vision-guided robotic sorting arm: sizing, assembly and wiring, VEX V5 programmed in C++, YOLOv8 detection driven over a serial link;
- Measured results: 93.2% success rate and a 5.1 s average cycle time. Final-year project defended in 2026;

Projects

Coin-operated automatic water dispenser

- Dispensing control logic, payment handling, user interface;
- PLC control of a pneumatic cylinder and a single-phase motor;
- Stepper motor and servo motor control;
- SaaS shipped under the UNITECH name: GES Mali and TBB Fashion, live — Next.js, Supabase;

Additional Training

- Computing — Exper-Lab, before the baccalaureate;
- Electronics and embedded systems, then artificial intelligence applied to IoT — Orange Digital Center, during university holidays;
- University robotics (VEX V5) — MaliBots 2024 Challenge, DoniFab;
- Mobile phone and computer repair — CEDEX, February – April 2026;

I certify that the information above is accurate and verifiable.